

PREDICTION OF SHOCK INJECTION LEVELS USING NORMALIZED ENERGY REPRESENTATION OF THE SOURCE

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ABSTRACT

The mechanical shock levels induced by a source on a structure are inherently structure-dependent. Current prediction methods frequently rely on analogy and past experience. While experimental testing remains essential, it is subject to considerable uncertainties. The proposed methodology offers an approach to characterize a shock source as a Normalized Energy Spectral Density through a dedicated reference test. This shock source characterization can subsequently be applied to a structure of interest to predict the shock levels resulting from multiple injections of the characterized shock source. This methodology has several applications, which are: predicting shock source levels from specific shock sources; obtaining a more accurate estimate of the injection shock levels on a system for its preliminary design so that the sizing rules can subsequently be applied; comparing shock levels on a given system for different mechanisms; optimizing shock test means by predicting shock levels across configurations; or using the Normalized Energy Spectral Density as shock source modeling.

1. INTRODUCTION

The development of space systems must ensure reliable operations despite exposure to various environmental stresses. Mechanical shocks are one of them. The considered shocks arise from nominal system operations, excluding hypervelocity impacts such as space debris. They can originate from various sources, including launcher stage separations, spacecraft-launcher separations, or appendage deployments such as solar arrays or antennas, for example [1,2]. They may also result from collisions or drops during terrestrial transport of the system [3,4] or from system landings or splashdowns [2]. Furthermore, taking the example of multiple separation devices that all have the same purpose—locking, releasing, and separating [5]—each will have its own shock levels depending on the design and characteristics.

To maintain the integrity and performance of sensitive components—such as electronic boards

and optical payloads—it is critical to accurately assess the system's resilience to anticipated flight shock events. Inadequate consideration may lead to performance degradation, underscoring the necessity of precise prediction and validation during the design phase. Currently, shock predictions are based on analogies with previous missions and empirical experience. However, the characterization of shock sources, the specifications of subsystems, and the qualification of components often depend on costly, risky, and poorly controlled tests. These tests introduce uncertainties, potentially resulting in under- or over-qualification due to inadequate representation of complex shock sources and their interactions with intricate structural systems.

The Shock Response Spectrum (SRS) [2,6–10] is commonly used to specify shock levels. For launcher-induced shocks, including separation events, launch service providers establish specifications at the launcher interface, with documented standards for vehicles such as Ariane 5 [2,11], Ariane 6 [12], Vega-C [13], Soyuz [2], and Falcon 9 [14], which are replicated in system-level tests. Besides, internal spacecraft shocks induced by appendage deployments, for example, require characterization through system-level or subsystem-level testing [2]. However, when testing subsystems or equipment, an accurate estimation of the interface shock levels is critical to ensure the representativeness of the test. Although methodologies for sinusoidal, random, and acoustic environments are well-established, mechanical shock prediction for space systems remains less mature, though progress is being made [2].

Numerical methods show promise, but require robust engineering methodologies to ensure accuracy for shock level prediction [15]. Among the numerical methods used for prediction in the field of mechanical shocks, the following can be cited: Finite Element Method (FEM), with explicit [16] or implicit [17] time-domain resolution, Statistical Energy Analysis (SEA) [18,19] and its hybridizations [20]. Other approaches, sometimes complementary to the previous methods, have been published, such as the work of Wang *et al.* [21] on Frequency Response Functions (FRF) or Zhao *et al.* [22]. However, current approaches in the field of mechanical shocks still rely on empirical sizing rules [9] and similarity methods based on previous projects [2].

Regarding the approaches presented previously, all require modeling of the shock source. Different methodologies exist for this modeling, but in all cases, one of the objectives of shock analyses is to determine the SRS levels at points of interest in structures, particularly at the source. This article presents a generalizable methodology for estimating shock levels at the shock source location. The proposed methodology is divided into two phases. Firstly, the Normlized Energy Representation of the source (NERS) approach aims at characterizing a shock source from an SRS, obtained experimentally on a reference specimen, in the form of a Normalized Energy Spectral Density (NESD). Subsequently, it can be transferred to another specimen to predict the expected shock levels when injecting one or more equivalent shock sources into the one characterized.

The article is organized as follows. First, Section 2 introduces the proposed methodology, presenting the overall process, without delving into specifics, and its potential applications. Then, Section 3 presents the various numerical tools used and the detailed methodology, including shock definitions and Shock Response Spectrum (SRS) specifications (Section 3.1); the computation of Energy Spectral Density from an experimentally obtained SRS (Section 3.2); the computation of the modal density weighted function by Kernel Density Estimation (KDE) methodology (Section 3.3); the Energy Spectral Density normalization (Section 3.4); and the overall process (Section 3.5).

Finally, having established the comprehensive set of computational tools, methodological conventions, and the overarching methodology, the article concludes with Section 4 by giving examples of the methodology applications, including ball drop tests with various masses and heights on different plates at multiple locations, as well as characterizations of different PyroShockGun (PSG) test configurations for injection shock level prediction on satellite structural models.

2. METHODOLOGY

In detail, the proposed methodology, illustrated in Figure 1, begins with step 0, obtaining a representative SRS of the injected shock levels from a characterized shock source on a reference specimen (see Section 3.1). Using this SRS and the total mass of the reference specimen, step 1 is the computation of an Energy Spectral Density (see Section 3.2).

Modal analysis enables the computation of a continuous frequency-based modal density function using Kernel Density Estimation (KDE) [23,24]. KDE is used, for example, in population density estimation and surface reconstruction from point clouds [25], but also for multimodality investigation [26] and density mode estimation [27]. Combined with modal weighting [28], the KDE is adapted for the proposed methodology in Section 3.3. This computation, step 2 of the methodology illustrated in Figure 1, results in a function that represents the number of modes per frequency band for a given impact location. This modal density weighted function is then used for step 3 to normalize spectral energy density, producing a Normalized Energy Spectral Density (NESD) for shock source characterization (see Section 3.4). The NESD can then be employed to estimate shock levels on a studied specimen by performing the inverse process (see Section 3.5).

For a studied specimen with multiple identical shock sources, the reverse processes of steps 2, 3, and 1 (denoted as steps 2', 3', and 1') can be applied to retrieve the SRS of the studied specimen. Modal analysis of the studied specimen combined with the NESD (scaled by the number of sources) enables the computation of injection Energy Spectral Density while applying energy conservation principles. Then the predictive SRS of injection levels can be recovered from the energy spectral density. When experimental data are available, numerical predictions can be compared with experimental results. The overall methodology is presented in Section 3.5.

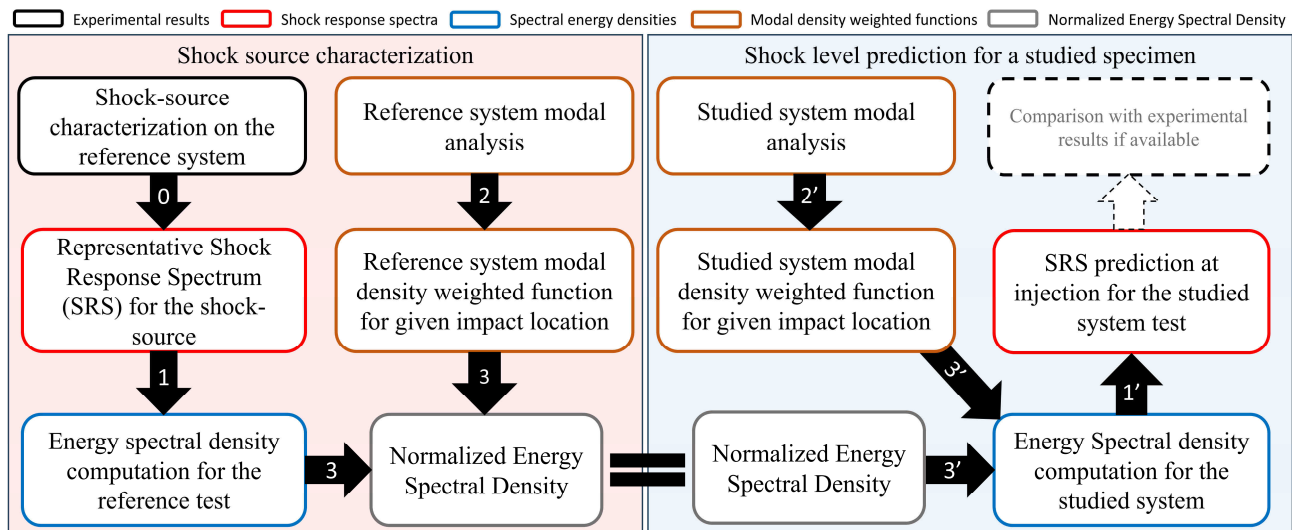


Figure 1 - Flowchart of the proposed methodology.

This methodology presents several applications in system development. In preliminary phases, it allows enhancing the precision of injection specifications for shock-sizing rules. One can characterize a shock source and estimate the shock levels expected on the studied specimen. This approach eliminates the need for a representative test, allowing appropriate injection level specifications to be defined during the early sizing phases with the estimation of injection shock levels. Additionally, for a studied specimen with multiple real shock-source variants (e.g., mechanisms, HDRMs), characterizing the sources and estimating the resulting shock levels on the structure enables efficient pre-selection of suitable shock-source designs without conducting tests on the structure itself. Furthermore, the normalized shock-energy representation, described by a Normalized Energy Spectral Density (NESD), can be used as a shock source model in simulation. The final example, and the examples given later, when considering various possible shock-test means or configurations for a specimen under study, characterizing the sources and estimating the shock levels on the structure allows for the pre-selection of the appropriate configuration of tests. This ensures that the required test specifications can be reproduced in advance.

At CNES, PyroShockguns (PSG) -- highly configurable shock generators that operate simultaneously -- are mounted directly on structures, including large spacecraft, to generate controlled mechanical shocks. The PSG operates by igniting a powder cartridge to propel an impactor, which strikes a damper at the end of the breech. The shock response exhibits significant variability based on the structural characteristics of the tested specimen, which complicates predictive modeling. This method would streamline the pre-determination of optimal parameters (damper materials, thicknesses, impactor travel distance, powder charge, number of shock sources) tailored to specific specimen requirements and specifications.

Ultimately, the proposed methodology can be applied to any shock source, in various domains. This generalization is feasible provided that the shock source can be characterized in another specimen, and that the modal density weighted function can be determined for the shock location in both the reference specimen and the specimen under study. The methodology maintains its assumptions across different applications as long as these fundamental conditions are satisfied.

The ensemble of tests conducted is performed for space applications and then are carried out under free-free boundary conditions. This is typically achieved by suspending the test specimen or placing it on foam supports. Experimental results obtained during experimental research at CNES have demonstrated that these boundary conditions effectively replicate the desired free-free conditions.

3. COMPUTATIONS TOOLS

3.1. Definition of a Shock and SRS Specification - Step 0

A mechanical shock, as defined in the ECSS handbooks [2,6], is a short-duration (50 μ s–20 ms), high-frequency (100 Hz–100 kHz) transient load of significant amplitude with an exponentially decaying envelope. It generates tensile–compressive, bending, and shear waves, along with reflection, dissipation, and diffraction phenomena. The loading phase lasts 50 μ s–20 ms, followed by a free-oscillation phase of up to 100 ms. While the time-domain signal provides useful information on peak amplitudes and event duration, it does not readily indicate the relative severity across the broad frequency content of a shock. Shock events typically contain short-duration high-frequency components and longer-lasting low-frequency structural responses. Frequency-domain analysis is therefore essential for accurately assessing shock severity and potential impact. For this reason, the Shock Response Spectrum (SRS) [9] is the widely adopted method for shock evaluation.

The SRS method [1] involves defining a set of single-degree-of-freedom (SDOF) systems, each characterized by its own natural frequency. The time-domain signal under study is applied as input to all of these SDOF systems. For each system, the absolute maximum temporal response is extracted and plotted against its natural frequency. In this case, acceleration time signals are used, and the process computes the absolute maximum temporal response of each SDOF with a given damping coefficient Q . This produces spectra that are easily comparable, unlike raw time-domain acceleration data. The SRS thus represents the shock severity across a studied frequency range. It is important to note that for a given time-domain signal, there is only one corresponding SRS. However, for a given SRS, there is an infinite number of time-domain signals that could produce it. Various SRS computation algorithms exist. While Kelly and Stricklin's method [7] is one example, the approach used in the present work follows Smallwood's method [8]. In the presented work, comparisons between numerical and experimental results will be accompanied by percentage values within the intervals of ± 3 , ± 4 , and ± 6 dB from experimental results, as is customary in terms of shock tolerances [29] and a correlation coefficient, which ranges from 0 to 1, which is the covariance between the numerical and experimental SRS divided by the product of their standard deviations.

Step 0 in the presented methodology in Figure 1 requires that the SRS computed from the experimental accelerations during tests must represent the SRS of the accelerations measured closest to the shock source.

3.2. Frequency-Domain Energy Spectral Density Computation from an SRS – Step 1

The purpose of this section is to construct, from an SRS alone, an energy distribution as a function of frequency. To achieve this, an upper bound on the maximum energy for each frequency is conventionally computed. For every computation frequency, the bounded energy corresponds to that of a single-degree-of-freedom (SDOF) system whose natural frequency equals the computation frequency and that is subjected to an excitation so that the peak acceleration of the SDOF matches the SRS value at that same frequency.

If the SRS is computed for N_f single-degree-of-freedom systems with mass, stiffness, natural frequency, and maximum acceleration $\{m_k, k_k, f_k\}_{k=1, \dots, N_f}$, the excitation that enables an SDOF of frequency f_k to reach the value $SRS(f_k)$ is identified, and the associated maximum energy is computed—to conventionally express a maximum energy as a function of frequency—in (Eq. 1).

$$E_{max}(f_k) = \frac{1}{8\pi^2} \frac{m_k \times SRS^2(f_k)}{f_k^2} \quad (\text{Eq. 1})$$

For a time-domain shock-acceleration signal examined with a sliding cumulative SRS, i.e., the Shock-Response Spectrum is recomputed at discrete time steps $t_i = i\Delta t$ on the signal segment that extends from the start $t = t_0$ to each t_i , it can be observed that initial high-frequency transients [2,6] produce early high-frequency SRS maxima. As modal behavior emerges, SRS maxima at lower frequencies are reached. This implies that the peak accelerations defined in the SRS do not occur simultaneously. Then, because of the non-simultaneous maxima, the effective mass for the mode associated to f_k can only be bound by total system mass denoted M_{tot} throughout the study. The bounded maximum energy is expressed as:

$$E_{sup}(f_k) = \frac{1}{8\pi^2} \frac{M_{tot} \times SRS^2(f_k)}{f_k^2} \quad (\text{Eq. 2})$$

To express this in terms of spectral density, considering the frequency band $[b_{k-1}; b_k]$ around the central frequency f_k , the Energy Spectral Density is written as follows:

$$ESD_{sup}(f_k) = \frac{1}{8\pi^2} \frac{M_{tot} \times SRS^2(f_k)}{f_k^2 (b_k - b_{k-1})} \quad (\text{Eq. 3})$$

(Eq. 3) represents the step 1 of the methodology presented and illustrated in Figure 1. A frequency discretization example used for large frequency range events is generally expressed as a geometric distribution of central frequencies $\{f_1, \dots, f_{N_f}\}$ such as $f_{k+1} = r f_k \equiv f_1 r^k$ with $r = 2^\Gamma$, the b_k values are defined as $b_0 = \frac{f_1}{r^{1/2}}$ and $b_k = \sqrt{f_k f_{k+1}} \equiv f_k r^{1/2}$. In the presented work, $\Gamma = \frac{1}{24}$, which corresponds to a frequency discretization of $\frac{1}{24}$ octave. For illustration, Figure 2 shows the SRS and the corresponding

computed energy for a 10.7 kg specimen across frequencies from 100 Hz to 10 kHz.

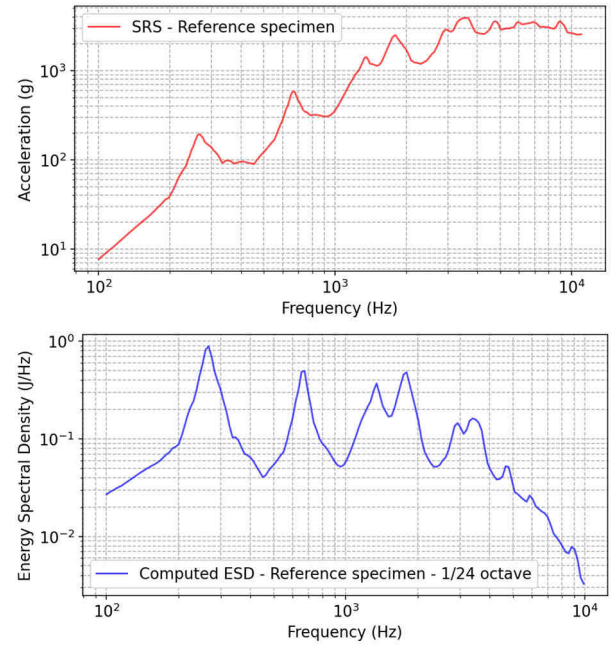


Figure 2 - A Shock Response Spectrum (SRS) and its associated maximum Energy Spectral Density for a frequency range of 100 Hz to 10 kHz, a 1/24 octave discretization $\Gamma = \frac{1}{24}$, and for a specimen with a total mass of 10.7 kg.

3.3. Computation of the Modal Density Weighted function – Step 2

The goal is to build a continuous frequency-dependent function representing the density of natural frequencies within a representative frequency range for a given impact zone and adapted for the normalization of the calculated energy spectral density. To quantify the distribution of natural frequencies in a structural system, we employ Kernel Density Estimation (KDE) [23], a non-parametric statistical method for estimating probability density functions.

At first, the kernel function is introduced and its relevance explained. Subsequently, the specific kernel function employed to adapt the modal density computation to the presented methodology is detailed. The last part provides a detailed explanation of the modal density computation, focusing on the modal weighting of the system's natural frequencies based on the modal amplitudes at the node located at the shock source, for a given direction.

Consider a system with N_{f_0} natural frequencies $\{f_{0i}\}_{i=1, \dots, N_{f_0}}$, determined through modal analysis.

Furthermore, consider again the set of $\{f_k = 2\pi\omega_k\}_{k=1, \dots, N_{f_k}}$ defining the frequency domain over which all quantities are evaluated. We propose a formulation for the weighted model density function $n(f_k)$ (the function is then normalized).

$$n_w(f_k) = \frac{1}{N_{f_0}} \sum_i K(f_{0i}, f_k) * W_{n,i}^{(d)} \quad (\text{Eq. 4})$$

As explained previously, our suitable kernel function $K(f_{0,i}, f_k)$ must be chosen such that the shape factors around the peaks in the modal density weighted function are consistent with those of the spectral energy density. To achieve this, our strategy relies on the relationship between the potential energy of the spring and the imposed acceleration.

Considering a mass-spring-damper model and computing the relative energy through the spring's potential energy, one can define a function that expresses, in the Fourier domain, the ratio of this energy to the square of the mass's acceleration. Once normalized, this yields the following:

$$K(f_{0,i}, f_k) = \frac{1}{\pi Q} \frac{f_{0,i}^3}{(f_{0,i}^2 - f_k^2)^2 + (f_{0,i} f_k / Q)^2} \quad (\text{Eq. 5})$$

For the time being, $Q_i \equiv Q$ is taken as a constant—identical for every mode—to keep the methodology straightforward. Nevertheless, one could envisage a more general formulation in which a separate Q value is assigned to each individual natural mode.

Experimentally, the shock source will likely excite certain modes more than others. Consequently, in the experimental SRS of the system under study, only peaks at the likely-excited modes will appear. Therefore, the natural frequencies of the system for the computation of the modal density weighted function, determined through modal analysis, must be weighted so that modes likely to be excited, relative to the shock impact zone along the excitation axis, contribute more to the computation of the modal density weighted function than those that are not. This methodology particularly allows differentiation based on the injection zone of the shock source.

Regarding the literature on modal density computation methodologies, particularly mode weighting, several definitions of mode participation have been proposed. Some combine effective mass [30,31], while others rely on observability factors [31,32]. Other approaches use Frequency Response Functions (FRFs) [33], and some rely on experimental methods to directly estimate the modal density [34,35]. In this work, the effective mass is excluded because it is not directly applicable under free-free boundary conditions, as the effective mass is concentrated on rigid-body modes. Some methods, such as the Effective Interface Mass approach [36], do attempt to define an effective mass for free-free systems.

Experimental identification is also excluded, as the objective is to avoid physical testing of the specimen studied. The observability factor defined in the Primodal user's guide [37] is also not retained. This factor, normalized by the maximum mode shape displacement across all degrees of freedom at each frequency, only indicates whether a given degree of freedom dominates others for a particular natural frequency; it does not distinguish observability differences between modes. It should be noted that there are various methodologies for defining modal participation factors [28], including domain-specific mathematical formulations [38–40]. In the proposed

approach, modal weighting is applied using a modal energy fraction. To weigh the modes in a multi-dimensional material system in space $\{X, Y, Z\}$, we limit our consideration to the discrete approximations of the material systems, based on the Finite Element Method (FEM). The latter allows for the computation of nodal displacements. The methodology presented introduces a modal participation factor $W_{n,i}^{(d)}$ within modal density weighted function computation, representing the fraction of modal energy associated with the displacement translation of the FEM discretization node n in the space axis direction $d \in \{X, Y, Z\}$ [28] such as:

$$W_{n,i}^{(d)} = \frac{|\phi_{ni}^{(d)}|^2}{\sum_{i=1}^N |\phi_{ni}^{(d)}|^2} \quad (\text{Eq. 6})$$

Here, $\phi_{ni}^{(d)}$ is the component of the mode shape vector for mode i at node n in direction (d) , $|\phi_{ni}^{(d)}|$ is the absolute amplitude of this component and $W_{n,i}^{(d)}$ is the modal energy fraction. That last one is expressed between 0 and 1 such that when it is close to 1, the mode is highly likely to be excited by an impact at this location and in direction (d) and vice versa. The modal participation factor $W_{n,i}^{(d)}$ is defined with a normalization with respect to the sum of the squares of the magnitudes of all amplitudes rather than the maximum magnitude of amplitudes per mode to preserve comparison between modes [28].

The normalization of weighting factors is merely conventional. In any case, the modal density weighted function is normalized subsequently as follows:

$$n(f_k) = N_{f_0} \frac{n_w(f_k)}{\sum_{k=1}^{N_{f_k}} n_w(f_k)} \quad (\text{Eq. 7})$$

The (Eq. 7) with the kernel function of (Eq. 5) and the weighting of modes in (Eq. 6) represents the step 2 of the methodology presented and illustrated in Figure 1. Note that for the presented work, modal analyses were performed using NASTRAN software. An illustration of the modal density weighted function for plate 1, with characteristics presented in Table 1, is given in Figure 3.

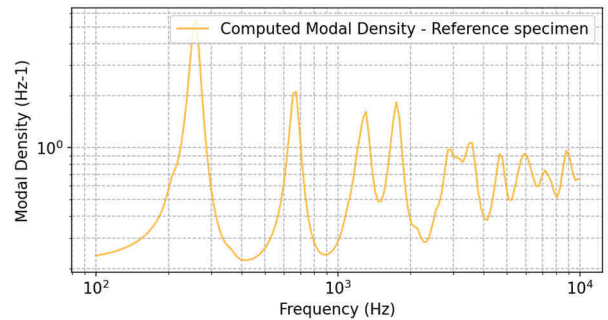


Figure 3 - Example of the modal density weighted function of plate 1 (see Table 1) for an impact at the center of the plate.

3.4. Energy Spectral Density Normalization - Step 3

In this section, we demonstrate the existence of a method to normalize energy in order to transfer this data from one system to another, thereby modeling a common source.

According to the works published by Cremer [41], or by Richard H. Lyon [42], and reused in works such as the thesis by Le Bot [43], the power input to a system, averaged over source location, excited by a noise excitation with a flat spectrum S_l over the band $\Delta\omega$, is expressed as follows:

$$\langle \Pi \rangle = \frac{\pi S_l \Delta\omega}{2} \frac{n(\omega)}{2\pi M} \quad (\text{Eq. 8})$$

Here, $n(\omega)$ is the modal density of the system and M its mass. In this manner, if we consider two systems 1 and 2 such that System 1 with mass M_1 and modal density $n_1(\omega)$ has undergone characterization of the impact source and received an average power $P_1(\omega)$ with spectrum $S_{l1}(\omega)$ over the interval $\Delta\omega$. System 2 with mass M_2 and modal density $n_2(\omega)$, must receive an average power $P_2(\omega)$ with spectrum $S_{l2}(\omega)$ over the interval $\Delta\omega$.

Then:

$$P_j = \frac{\pi S_{lj}(\omega) \Delta\omega}{2} \frac{n_j(\omega)}{2\pi M_j}, \quad j \in \{1, 2\} \quad (\text{Eq. 9})$$

By posing $P_{norm,j} = \frac{P_j(\omega) M_j}{n_j(\omega)}$, then if $S_{l1}(\omega) = S_{l2}(\omega)$,

so $P_{norm,1} = P_{norm,2}$.

Through the energy-power relationship in the Laplace domain, and by dividing by the frequency band, we can then define a normalization by energy, in the form of a spectral energy density, such that The Normalized Energy Spectral Density (NESD) can be expressed as follows:

$$NESD = \frac{ESD(\omega) M_j}{n_j(\omega)}, \quad (\text{Eq. 10})$$

respecting the same conclusions. This formulation of a normalized spectral energy density then allows us to transfer this data from one system to another by considering that the shock source is identical. The use of spectral energy density rather than energy allows us to disregard differences in the frequency discretization definition of the reference case data and the desired output data. The NESD is conventionally defined by analogy to the normalization process used for direct and reverberant energy fields in Statistical Energy Analysis (SEA), where energy densities are expressed in terms of modal density [2,44,45].

3.5. Shock source normalization and reverse process for prediction

The methodology illustrated in Figure 1 computes a normalized energy spectral density (NESD) from the SRS of a shock source on a reference system and the modal analysis of the complete configuration test (i.e. the reference system and the shock source). Reversing the process, using the NESD and a modal analysis of a target specimen subjected to identical shock sources, a representative SRS can be predicted. The computation steps will be described subsequently.

The analysis involves the following:

- a set of N_{fk} frequencies $\{f_1, \dots, f_{N_{fk}}\}$ as the discretization of the set of defined quantities.
- a Shock Response Spectrum (SRS) $SRS_{ref}(f_k)$ representative of the levels at the injection of a shock source on a reference specimen with a mass M_{ref}
- a studied specimen with a mass M_{study} and N_s shock source(s) identical to the shock source characterized.
- a single damping coefficient value $Q = \frac{1}{2\zeta}$ applied uniformly across the entire frequency range.

For reference specimen,

- a set of natural frequencies $\{f_{0r,1}, \dots, f_{0r,N_{f0r}}\}$ for the reference specimen.
- a modal density weighted function $n_{ref}(f_k)$ from (Eq. 7) with the proper mode weighting $W_r(f_{0r,i})$. (Eq. 6) based on the contributions of the natural frequencies of the reference specimen to its behavior for the given shock source location.

Similarly, for the studied specimen,

- a set of natural frequencies $\{f_{0s,1}, \dots, f_{0s,N_{f0s}}\}$ for the reference specimen.
- a modal density weighted function $n_{study}(f_k)$ from (Eq. 7) with the proper mode weighting $W_s(f_{0s,i})$. (Eq. 6) based on the contributions of the natural frequencies of the reference specimen to its behavior for the given shock source location.

The first part of the methodology, comprising steps 1–3, is dedicated to characterizing the shock source using the reference specimen of mass M_{ref} .

Following the approach described in Section 3.2, step 1 of the methodology presented in Figure 1 is the computation of the maximum Energy Spectral Density for the shock source on the reference specimen is expressed using

$$ESD_{ref}(f_k) = \frac{1}{8\pi^2} \frac{M_{ref} \times SRS_{ref}^2(f_k)}{f_k^2 (b_k - b_{k-1})}. \quad (\text{Eq. 11})$$

Following the approach describes in Section 3.3, step 2 of the methodology is to compute $n_{ref}(f_k)$.

Following the approach described in Section 3.4, step 3 of the methodology is to compute the Normalized Energy Spectral Density (NESD) that can be expressed as follow:

$$NESD(f_k) = \frac{ESD_{ref}(f_k)M_{ref}}{n_{ref}(f_k)}, \quad (Eq. 12)$$

The analysis now considers the specimen under study with mass M_{study} . This specimen is intended to receive N_s shock source(s) identical to the shock source characterized.

Following the approach described in Section 3.3, step 2' of the methodology is to compute $n_{study}(f_k)$. Following the approach described in Section 3.4, step 3' of the methodology is to compute the Energy Spectral Density for the studied specimen using the NESD:

$$ESD_{study}(f_k) = \frac{NESD(f_k)n_{study}(f_k)}{M_{study}}N_s, \quad (Eq. 13)$$

In practice, due to the methodology for computing the modal density weighted function, energy conservation is not necessarily maintained during the transfer. An energy conservation factor is introduced to enforce energy conservation:

$$k_{cons} = \frac{\sum_{k=1}^{N_{fk}} ESD_{ref}(f_k)}{\sum_{k=1}^{N_{fk}} ESD_{study}(f_k)} N_s \quad (Eq. 14)$$

Thus, the conserved maximum energy spectral density can be expressed as:

$$ESD_{study,cons}(f_k) = ESD_{study}(f_k)k_{cons} \quad (Eq. 15)$$

Consequently, following the approach described in Section 3.2, step 1' of the methodology is the computation of the predicted SRS for the injection of N_s shock sources into the specimen under study is given by:

$$SRS_{study}(f_k) = \sqrt{\frac{8\pi^2 f_k^2 (b_k - b_{k-1}) ESD_{study,cons}(f_k)}{M_{study}}} \quad (Eq. 16)$$

For example, from the SRS illustrated in Figure 2, the Energy Spectral Density has been computed and illustrated as well in Figure 2; then the weighted modal density function has been computed and illustrated in Figure 3; and finally, the NESD can be computed and illustrated in Figure 4.

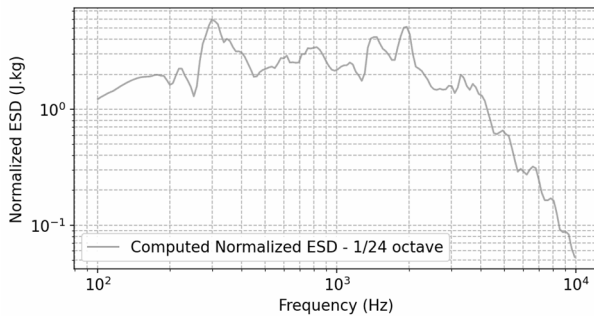


Figure 4 - Example of a NESD characterized for a 255g ball impact dropped at 15cm (frequency range 100 Hz to 10 kHz - 1/24 octave discretization).

4. APPLICATIONS

This section aims to apply the methodology to characterize a shock source on a reference system and to predict the injected shock level on a studied specimen. Plate and ball materials are detailed in Table 1, and specimens are illustrated in Figure 5 and Figure 6. Further complex specimens such as structural models of microsatellite are studied and illustrated in Figure 7.

Table 1: Material properties and experimentation mass characteristics for the ball impactor and the plate used in experimentation.

Material Name	Plate 1	Plate 2	Ball
Material	Al. 2618 T6	Aluminium	Stainless steel 304L
Density	2780 kg/m ³	2854 kg/m ³	7800 kg/m ³
Young's Modulus	74.5 GPa	72.0 GPa	195 GPa
Poisson Coef.	0,3	0,3	0,3
Size (mm ³)	543.5×560 × 13	440×400 × 10	
Mass	10,7 kg	5 kg	67g to 535 g

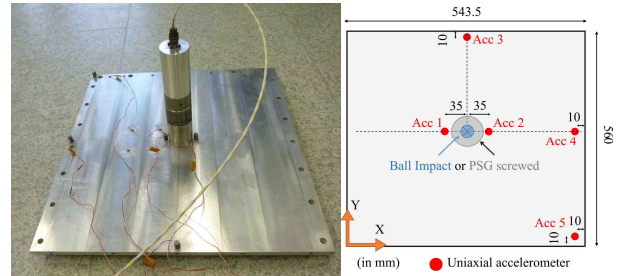


Figure 5 - On the left: Illustration of plate 1's instrumentation with the PyroshockGun shock source. On the right: scheme of the instrumentation of plate 1.

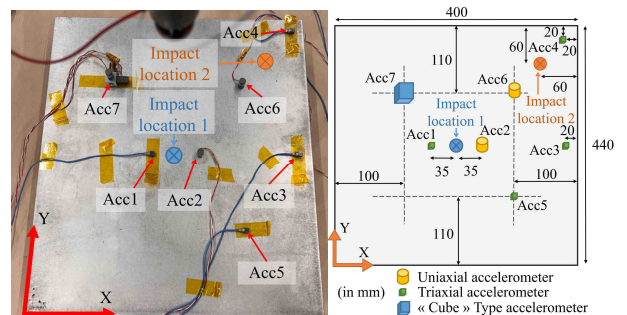


Figure 6 - On the left: Illustration of plate 2's instrumentation with the steel ball shock source, with two impact locations. On the right: scheme of the instrumentation of plate 2.

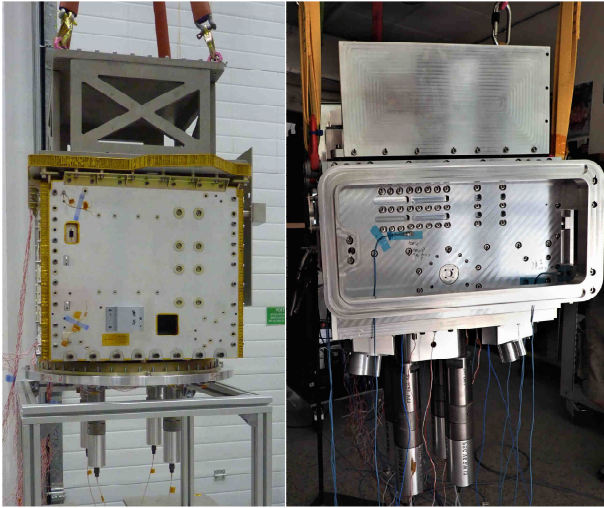


Figure 7 - On the left: Illustration of the MicroCarb Structural model with 4 PSG shock sources. On the right: Illustration of the Structural model of a microsatellite with 4 PSG shock sources.

4.1. Shock injection level prediction with NERS approach for ball drops on plates

In the following example, the same characterization is employed to predict two distinct shock levels applied at two different shock locations on the specimen under study.

- Figure 8: Characterization of a 255 g ball drop from 15 cm on Plate 1 providing the desired Normalized Energy Spectral Density (NESD).
- Figure 9: Use of the NESD from Figure 8 to predict shock injection level on Plate 2 for a **centered** drop.
- Figure 10: Use of the NESD from Figure 8 to predict shock injection level on Plate 2 for a **corner** drop.

Further examples can be presented using the characterization of a 255 g ball dropped from 60 cm and a 535 g ball dropped from 30 cm onto Plate 1. These data are then used to predict the corresponding shock-injection levels for centered drops on Plate 2. Figure 11 illustrates two points. First, the Normalized Energy Spectral Density (NESD) exhibits a higher-amplitude (more energetic) response than the NESD obtained with the 255-g ball released from a height of 15 cm. Second, the weighted modal-density function is identical to that shown in Figure 9, because both figures pertain to the same specimen and the same source location. Finally, through this methodology we consistently obtain results that are coherent between prediction and experimentation when looking at the SRS results.

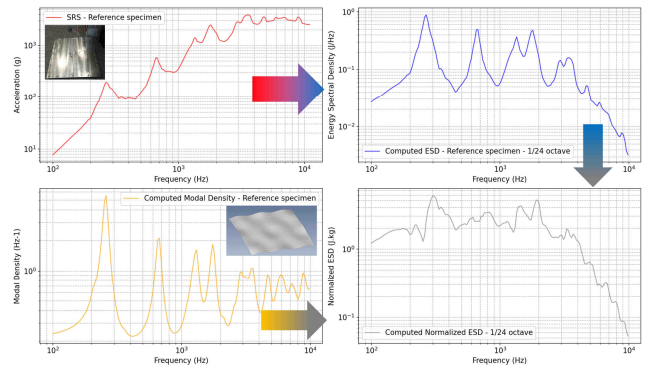


Figure 8 - 255 g ball drop from 15 cm on Plate 1 (center) – characterization.

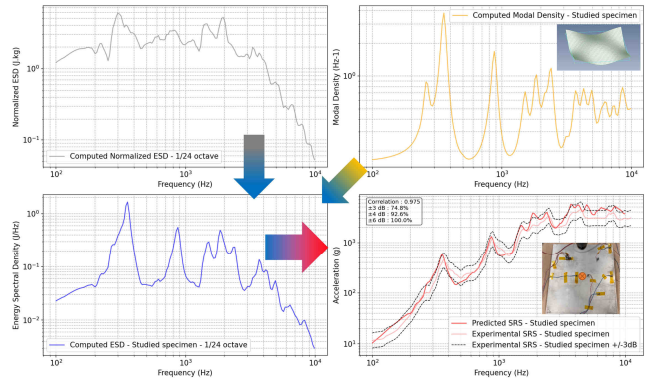


Figure 9 - Prediction for 255 g ball drop from 15 cm on Plate 2 (center) considering characterization illustrated in Figure 8.

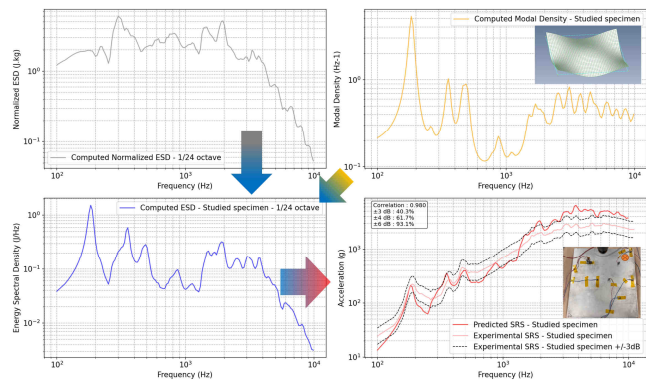


Figure 10 - Prediction for 255 g ball drop from 15 cm on Plate 2 (corner) considering characterization illustrated in Figure 8.

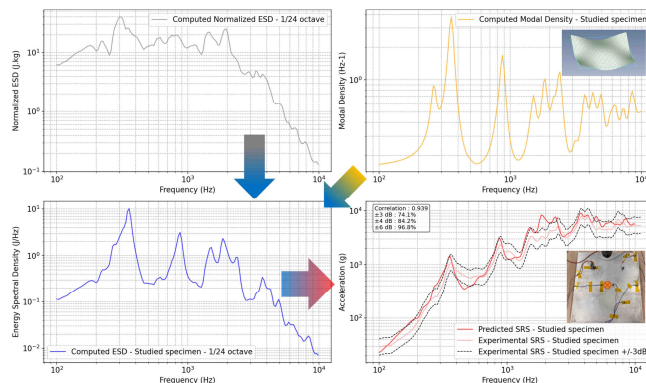


Figure 11 - Prediction for 535 g ball drop from 30 cm on Plate 2 (center) considering its (unillustrated) characterization.

4.2. Shock injection level prediction with NERS approach for concrete applications and operational outcomes

The subsequent examples present concrete engineering applications of the NERS approach. Several application contexts can be considered:

- **Enhanced precision of injection specifications for shock-sizing rules in preliminary analyses:** consider a studied specimen and an actual impact source. By characterizing the source and estimating the shock levels imparted to the structure, a representative test is no longer required; instead, appropriate level specifications can be defined for injection during the early sizing phases.
- **Pre-selection of the design for shock sources** (e.g., mechanisms, HDRMs, ...): consider a studied specimen and several real shock-source variants. Characterizing the sources and estimating the resulting shock levels on the structure enables efficient identification of appropriate shock-source designs without conducting tests on the structure itself.
- **Pre-selection of the shock-test configuration:** consider a studied specimen and several possible shock-test setups. By characterizing the sources and estimating the shock levels on the structure, the appropriate configuration of shock sources can be identified in advance to reproduce the required test specifications.
- **Use of a normalized shock-energy representation for source simulation:** the characterization step and its description by a normalized spectral energy density enable the shock source to be represented in a simulation independently of the modeled structure.

In this section, examples of characterizations of the CNES shock test mean PyroShockGun are given for the purpose of determining the firing configurations needed to reproduce the prescribed SRS specifications on the satellite structural models studied. By first characterizing the PSG on a reference specimen, we were able to predict the SRS levels that would be injected into satellite models. These examples extend the validation to complex structures as for a full satellite structural model, thereby demonstrating the scalability and versatility of the proposed method for realistic applications.

4.2.1. ECY

For example, consider a satellite structural model that will be sent to a test campaign, illustrated on the right in Figure 7. The shock test means at CNES are primarily PSGs, whose configurations must be determined to reproduce the required test specifications. By conducting a characterization campaign of the PSG in various configurations on a reference structure—as shown in Figure 5—and having the modal analyses of both the reference tests and the SM, the methodology allows prediction of the injection levels that will be experienced in tests on the SM.

Consider the characterization performed for one “high-shock” configuration of the PSG. Using the experimental acceleration data from accelerometers 1 and 2 located near the source, and by computing a weighted modal-density function, a reference NESD is obtained for all subsequent predictions for this PSG configuration as illustrated in Figure 12.

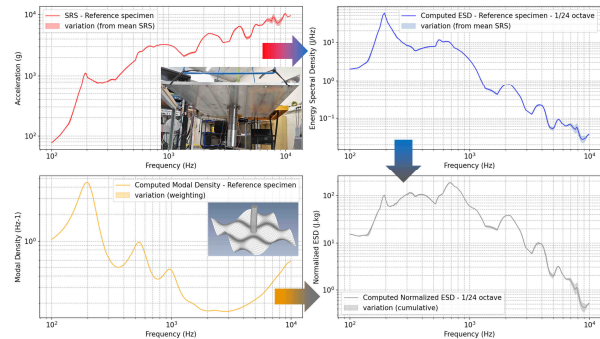


Figure 12 - ECPSC 2026: NERS approach for shock source characterization. Characterization of the PSG on the suspended EC1 plate.

Based on this characterization of the PSG in its firing configuration, the methodology can be used to predict shock levels for the structural model with the same firing configuration but potentially in different positions, as illustrated in Figure 13.

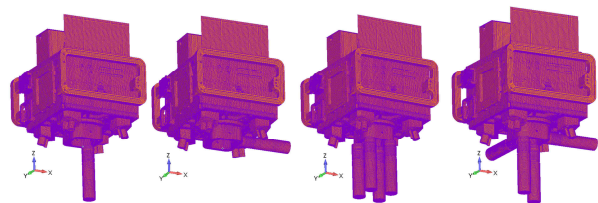


Figure 13 - Firing configurations for the ECY campaign.

Consequently, the use of the NERS approach is presented below for predicting shock levels on the structural model:

- for a PSG positioned along the Z-axis (out-of-plane), refer to Figure 14.
- for a PSG positioned along the X-axis (in-plane), refer to Figure 15.
- for four PSGs positioned along the Z-axis (out-of-plane), refer to Figure 16.
- for four PSGs with two on the Z-axis, one on X and one on Y, refer to Figure 17 focusing on the Y axis.

Further examples can be presented for the multi-axial shock test and with another PSG configuration.

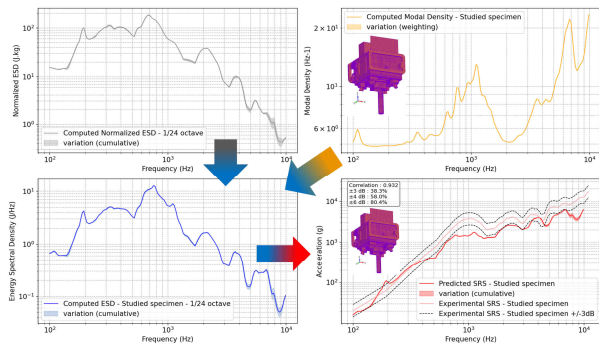


Figure 14 - ECY 2026: NERS approach for shock level prediction with one PSG in Z axis. Comparison between prediction and experimental results.

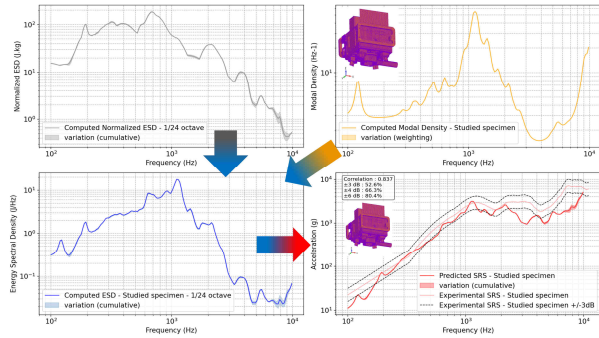


Figure 15 - ECY 2026: NERS approach for shock level prediction with one PSG in X axis. Comparison between prediction and experimental results.

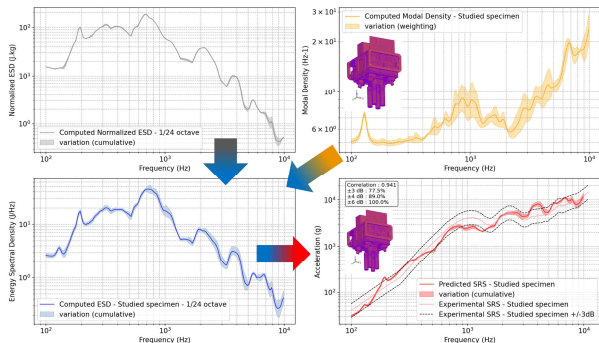


Figure 16 - ECY 2026: NERS approach for shock level prediction with four PSG in Z axis. Comparison between prediction and experimental results.

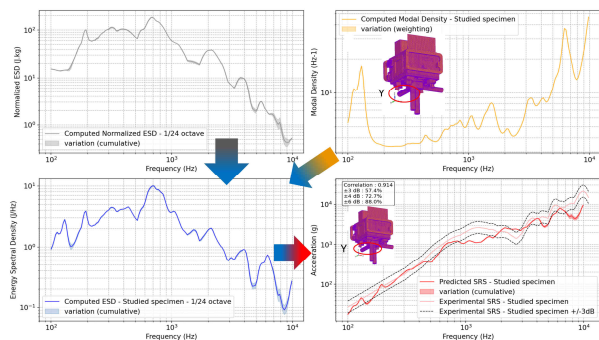


Figure 17 - ECY 2026: NERS approach for shock level prediction with four PSG in X,Y and Z axis, considering the Y axis. Comparison between prediction and experimental results.

4.2.2.EC MicroCarb SM

For example, consider the MicroCarb structural model that will be sent to a test campaign, illustrated on the left in Figure 7. The PSG configuration must be determined to reproduce the required test specifications. Considering 3 (unillustrated) characterizations of 3 different configurations of the PSG—as illustrated in Figure 5—categorized as “low shock”, “medium shock” and “high shock”, we obtain the shock source levels prediction in Figure 18, Figure 19 and Figure 20.

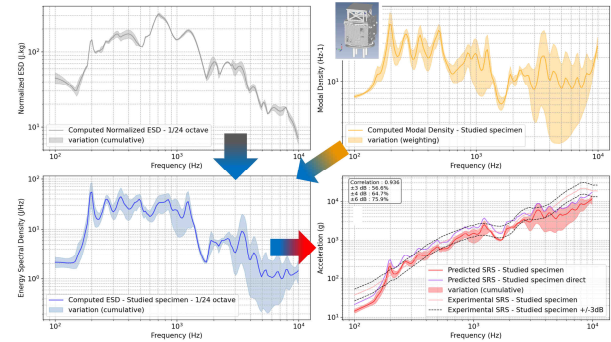


Figure 18 - ECSM 2020: NERS approach for shock level prediction with four PSG in Z axis in a “low-shock” configuration, considering the Z axis. Comparison between prediction and experimental results.

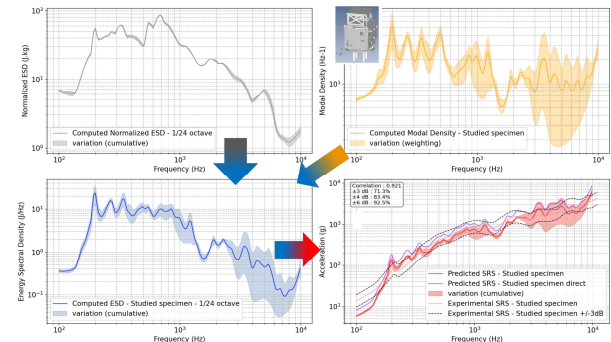


Figure 19 - ECSM 2020: NERS approach for shock level prediction with four PSG in Z axis in a “medium-shock” configuration, considering the Z axis. Comparison between prediction and experimental results.

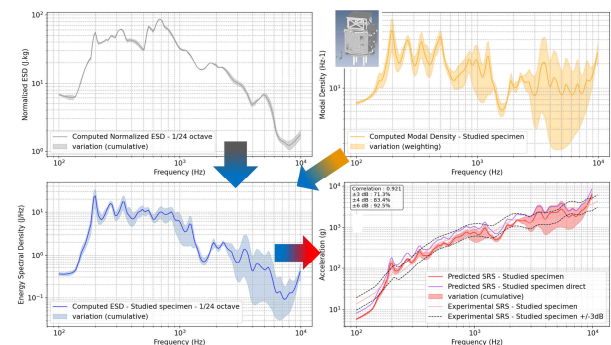


Figure 20 - ECSM 2020: NERS approach for shock level prediction with four PSG in Z axis in a “high-shock” configuration, considering the Z axis. Comparison between prediction and experimental results.

4.3. Empirical demonstration of the normalized shock source representation through the previous examples

The way to demonstrate the effectiveness of the methodology is to characterize the same shock source on different specimens, with distinct source locations and along various measurement axes. Experimental results show that the computed SRS depend entirely on the impacted structure and on the position of the source (unillustrated but demonstrated with the previous examples). In contrast, the figures presented below reveal a marked similarity among the Normalized Energy Spectral Densities (NESDs) obtained for these differing configurations.

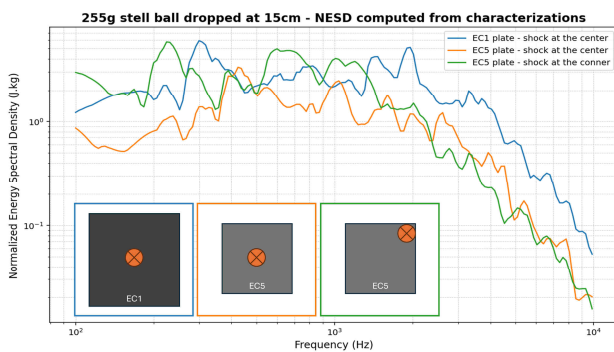


Figure 21 - Comparison of NESDs Characterized During the Ball-Drop-onto-Plate Campaign.

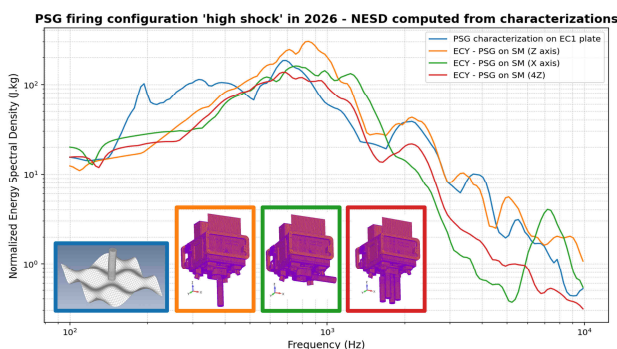


Figure 22 - Comparison of the NESDs characterized in the ECY campaign for a "high shock configuration".

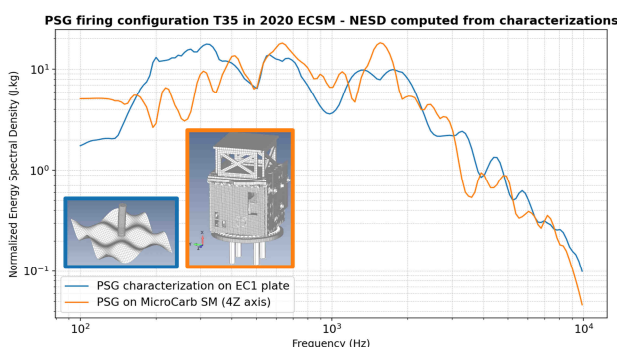


Figure 23 - Comparison of the NESDs characterized in the ECSM campaign for a "low shock configuration".

Several of the peak-like discrepancies can be traced to the simplifying assumption in the methodology of assigning a Q damping value to all natural modes. This uniform damping prevents a fitting normalization of the peaks observed in the experimental SRS, because those peaks are governed not only by the Q damping prescribed in the SRS formulation but also by the intrinsic damping of the structure itself.

5. CONCLUSIONS

In conclusion, this study introduces a novel methodology that **reliably represents a shock source through a normalized representation based on the Normalized Energy Spectral Density**, computed from a characterization test on a reference specimen. This NESD is then used for shock source levels prediction on a studied specimen subjected to the same shock sources as characterized. This approach not only achieves the objectives to **enhance the characterization of shock sources** and obtain **usable data for shock-source modeling in simulation** for example, but it also allows a **more accurate estimation of shock source level** for preliminary design sizing and application of ECSS rules, for example, and permits **prediction of injection shock levels, comparison and optimization for different configurations**, including **shock test means** by pre-defining optimal parameters tailored to a given specimen's specifications, **or mechanisms**.

The methodology highlighted several key aspects that prove to be essential to its effectiveness: using a representative computed SRS of the shock source levels from measurements in characterization, employing a kernel function specifically adapted to the methodology, defining modal weighting in the computation of the modal density weighted function, and the normalization of the energy spectral density by the weighted modal density function.

Finally, the versatility of this methodology lies in its ability to be applied across different domains as demonstrated with the given examples. This methodology represents a significant advancement in shock source modelling and in shock source level prediction, offering a more accurate, flexible, and efficient alternative to shock source level estimation and prediction for preliminary design phases or tests. Its implementation could streamline specimen development processes and improve the reliability of shock testing, ensuring better performance and safety in various engineering applications.

Improvements can be made in both determining and implementing distinct damping factors for each individual mode. Future work could also explore characterization based on alternative input data—such as a force—as a further avenue of investigation.

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